MAIN SYSTEM ARDUINO

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <RTClib.h>
#include <SoftwareSerial.h>
#include <DFRobotDFPlayerMini.h>
#include <RH_ASK.h>
#include <SPI.h>
#include <Servo.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
RTC_DS3231 rtc;

SoftwareSerial mySerial(10, 11);
DFRobotDFPlayerMini player;

RH_ASK rf_driver(2000, -1, 8, -1);

Servo myServo;
const int servoPin = 9;

const int buttonPin = 2;
const int irPin = 3;

unsigned long previousMillis = 0;
const long interval = 1000;
bool systemBusy = false;

void setup() {

  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Starting...");
  delay(1000);

  if (!rtc.begin()) {
    lcd.clear();
    lcd.print("RTC Error");
    while (1);
  }
```

```
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(irPin, INPUT);

  myServo.attach(9);
  myServo.write(0);

  mySerial.begin(9600);

  if (!player.begin(mySerial)) {
    lcd.clear();
    lcd.print("DF Error");
    while (1);
  }

  player.volume(20);

  if (!rf_driver.init()) {
    lcd.clear();
    lcd.print("RF Error");
    while (1);
  }

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("System Ready");

  delay(2000);
  lcd.clear();
}

void loop() {

  displayTime();

  if ((digitalRead(buttonPin) == LOW ||
       digitalRead(irPin) == LOW) &&
      !systemBusy) {

    activateSystem();

    sendRF("ON");
  }
}

void displayTime() {

  unsigned long currentMillis = millis();
```

```
  if (currentMillis - previousMillis >= interval) {

    previousMillis = currentMillis;

    DateTime now = rtc.now();

    lcd.setCursor(0, 0);
    lcd.print("Time:          ");

    lcd.setCursor(0, 1);

    if (now.hour() < 10) lcd.print("0");
    lcd.print(now.hour());

    lcd.print(":");

    if (now.minute() < 10) lcd.print("0");
    lcd.print(now.minute());

    lcd.print(":");

    if (now.second() < 10) lcd.print("0");
    lcd.print(now.second());

    lcd.print("   ");
  }
}

void activateSystem() {

  systemBusy = true;

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Activated!");

  lcd.setCursor(0, 1);
  lcd.print("Processing");

  player.play(1);

  myServo.write(90);

  delay(1000);

  myServo.write(0);
```

```
  delay(1000);

  lcd.clear();

  systemBusy = false;
}

void sendRF(const char *msg) {

  rf_driver.send((uint8_t *)msg, strlen(msg));

  rf_driver.waitPacketSent();

  Serial.print("Sent RF: ");
  Serial.println(msg);
}
```

RECEIVER ARDUINO

```
#include <RH_ASK.h>
#include <SPI.h>

RH_ASK rf_driver(2000, 8, -1, -1);

const int ledPin = 7;
const int vibratorPin = 9;

void setup() {

  Serial.begin(9600);

  if (!rf_driver.init()) {
    Serial.println("RF Init Failed");
    while (1);
  }

  pinMode(ledPin, OUTPUT);
  pinMode(vibratorPin, OUTPUT);

  digitalWrite(ledPin, LOW);
  digitalWrite(vibratorPin, LOW);

  Serial.println("Receiver Ready");
}
```

```
void loop() {

  uint8_t buf[50];
  uint8_t buflen = sizeof(buf);

  if (rf_driver.recv(buf, &buflen)) {

    buf[buflen] = '\0';

    String msg = String((char*)buf);
    msg.trim();

    Serial.print("Received: ");
    Serial.println(msg);

    if (msg == "ON") {

      digitalWrite(ledPin, HIGH);
      digitalWrite(vibratorPin, HIGH);

      delay(1000);

      digitalWrite(vibratorPin, LOW);
    }
    else if (msg == "OFF") {

      digitalWrite(ledPin, LOW);
      digitalWrite(vibratorPin, LOW);
    }
  }
}
```

CONNECTIONS

MAIN SYSTEM CONNECTIONS

LCD I2C

- SDA -> A4
- SCL -> A5
- VCC -> 5V
- GND -> GND

RTC DS3231

- SDA -> A4
- SCL -> A5
- VCC -> 5V
- GND -> GND

DFPLAYER MINI

- TX -> D10
- RX -> D11
- VCC -> 5V
- GND -> GND
- SPK1/SPK2 -> Speaker

433MHz RF TRANSMITTER

- DATA -> D8
- VCC -> 5V
- GND -> GND

SERVO MOTOR

- SIGNAL -> D9
- VCC -> 5V
- GND -> GND

BUTTON

- One Side -> D2
- Other Side -> GND

IR SENSOR

- OUT -> D3
- VCC -> 5V
- GND -> GND

RECEIVER CONNECTIONS

433MHz RF RECEIVER

- DATA -> D8
- VCC -> 5V
- GND -> GND

LED

- Positive -> D7
- Negative -> 220 Ohm Resistor -> GND

VIBRATION MOTOR

- Positive -> D9
- Negative -> GND